From the Superderivative to a Relative-Gradient Edge Operator

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***Abstract*—First-derivative edge detectors such as Sobel, Prewitt, Roberts and Scharr measure absolute grey-level differences. We revisit the log-gradient magnitude, a relative-gradient edge operator, and motivate it by analogy with the superderivative, a multiplicative derivative in which the independent variable is perturbed multiplicatively rather than additively. In one and two dimensions the logarithm of the superderivative is a radial logarithmic derivative; its local, origin-free, isotropic analogue is the magnitude of the logarithmic gradient. The resulting operator measures relative rather than absolute change, giving an edge-strength map that is invariant under global intensity rescaling. We present the theory, show how the operator can be implemented in two steps—a classical gradient kernel plus a pointwise division—and compare it with standard first-derivative filters on synthetic images with ground-truth edges and on the official BSDS500 benchmark. We also study a parametric family that interpolates between the absolute gradient and the log-gradient; empirically, the best members are small intermediate exponents, which edge the classical kernels by a small but statistically detectable margin in the per-image metric and, with the same Gaussian smoothing as the single-σ Canny, match it in ODS and lead it in OIS and AP, while the tuned Canny remains ahead in ODS. The single-threshold margin over the kernels does not replicate on the held-out test split, and the fully relative endpoint is never the best member, its ranking depending on the edge-map normalisation. The contribution is the construction and the invariance it buys, not a new state of the art.**

***Index Terms*—superderivative, edge detection, first-derivative filters, relative gradient, log-gradient, Sobel.**

I. Introduction

EDGE detection is one of the oldest and most studied tasks in image processing. Classical first-derivative operators — Sobel, Prewitt, Roberts, Scharr, and many others — estimate the spatial gradient of the image intensity and declare edges where the gradient magnitude is large. All of them share the same assumption: a grey-level jump of a given size is equally important anywhere in the image.

That assumption is questionable. A jump of ten grey levels on a background of intensity five is visually much stronger than the same jump on a background of intensity two hundred. The human visual system responds approximately logarithmically to light [1], so the relevant quantity is the *relative* jump, not the absolute one. Several authors have incorporated this idea into tone mapping, enhancement and denoising, usually by working on the logarithm of the image. Homomorphic filtering separates illumination from reflectance by taking logarithms before filtering and exponentiating afterwards — a multiplicative image model $I = R \cdot L$ is turned into an additive one by $\log I = \log R + \log L$, filtered, and mapped back [7]; Retinex theory builds lightness from ratios of neighbouring intensities, so that the recovered quantity is a relative rather than an absolute one [3], [4], and its later variational and centre/surround formulations keep that property [5], [6]; tone-mapping operators compress dynamic range on a logarithmic or otherwise relative scale [2], [25]. All of these act on the image before a derivative is taken. The operator proposed here puts the same relative-change principle into the derivative itself.

The *superderivative* is the derivative obtained by perturbing the independent variable multiplicatively rather than additively [9]; in multiplicative (non-Newtonian) calculus the same construction is the bigeometric derivative [8]. Whereas the ordinary derivative adds a small increment h to the argument, $x \mapsto x + h$, the superderivative multiplies the argument by a factor close to one, $x \mapsto x \cdot (1 + h)$. The resulting quantity therefore measures the *relative* growth of f with respect to a proportional change of its input, rather than the absolute growth with respect to an additive change. The ordinary derivative admits other generalisations as well: fractional calculus extends it to arbitrary real order through the Riemann–Liouville, Caputo and Grünwald–Letnikov operators [29]–[31], whereas the superderivative changes the perturbation itself. The two generalisations are independent and should not be confused: fractional calculus keeps the additive perturbation $x \mapsto x + h$ and makes the *order* real, while the superderivative keeps the order at one and makes the *perturbation* multiplicative; the superderivative is therefore *not* a derivative of arbitrary order, and its logarithm is an ordinary first-order derivative of $\log f$. We use this relative-change principle to motivate a family of first-derivative edge operators. The simplest member of the family is the log-gradient magnitude

$$E(x, y) = \|\nabla \log I(x, y)\| = \frac{\|\nabla I(x, y)\|}{I(x, y)}, \tag{1}$$

which replaces the absolute gradient with the relative gradient. More generally, the original two-dimensional superderivative score

$$L(x, y) = \frac{x\, \partial_x I(x, y) + y\, \partial_y I(x, y)}{I(x, y)} \tag{2}$$

measures relative change under a radial zoom centred at the origin.

Our contributions are:

- A construction motivating the log-gradient edge map as the local, isotropic, origin-free analogue of the superderivative, together with the two-dimensional radial score it comes from. The operator is motivated by, not derived from, the multiplicative derivative: the origin-dependent weights of the latter are what its isotropic form drops.
- A proof of invariance under global intensity rescaling.
- A parametric family $E_p = \|\nabla I\|/I^p$ that interpolates between the absolute gradient ($p = 0$) and the log-gradient ($p = 1$).
- A discussion of why the operator is not, in its full radial form, a pure convolution, and how it can nevertheless be evaluated in two efficient steps.
- Experimental comparisons with Sobel, Prewitt, Roberts, Scharr and Canny on synthetic ground-truth images, natural images, and BSDS500.

Applications beyond edge detection are outside the present scope.

II. Background: classical first-derivative kernels

All classical first-derivative operators estimate the same quantity, the spatial gradient, but use different finite-difference masks. Table I lists the most common 3×3 (and 2×2) kernels [14].

Several other designs exist. The *compass operators* of Kirsch [20] and Robinson [21] use a set of oriented 3×3 masks (typically eight directions) and take the maximum response as the edge strength. The Farid–Simoncelli filters use larger, separable 5×5 kernels that are optimally steerable and have better rotation invariance [22], [24].

The gradient magnitude is $E = \sqrt{(G_x * I)^2 + (G_y * I)^2}$, where $*$ denotes convolution. The Canny detector adds Gaussian smoothing, non-maximum suppression and hysteresis thresholding to a gradient estimate; it is a pipeline rather than a single kernel [23]. The scale-space framework that underlies multiscale edge detection is developed by Lindeberg [27]. All of these methods treat a given absolute grey-level jump as equally significant regardless of the local mean. Indeed, consider a one-dimensional step from a baseline I_0 to $I_0 + \Delta I$. The derivative is the same for any I_0 because $\partial_x(I_0 + \Delta I\, H(x)) = \Delta I\, \delta(x)$, where H is the Heaviside step and δ the Dirac delta. The measured edge strength is therefore proportional to $|\Delta I|$ and independent of I_0. Equivalently, at a fixed *contrast ratio* $\Delta I/I_0$ the absolute response grows with the background I_0, so a fixed threshold detects equal-contrast edges unequally; this is the regime in which the relative operator of Section IV differs from the classical one, and it is tested directly in Section VI-K.

III. The superderivative

A. One-dimensional definition

For a positive function f, the superderivative is defined by [8], [9]

$$\mathrm{SD}(f(x)) = \lim_{h\to 0} \sqrt[h]{\frac{f(x\cdot(1+h))}{f(x)}} = \exp\Big(x\,\frac{f'(x)}{f(x)}\Big). \quad (3)$$

The limit follows from the first-order expansion $f(x\cdot(1+h)) = f(x+xh) \approx f(x) + xhf'(x)$, which gives

$$\frac{f(x\cdot(1+h))}{f(x)} \approx 1 + xh\,\frac{f'(x)}{f(x)}. \quad (4)$$

Write $a = x\,\frac{f'(x)}{f(x)}$ and assume $a \neq 0$ (if $a = 0$ the limit is $1 = e^0$). Then

$$\Big(1 + xh\,\frac{f'(x)}{f(x)}\Big)^{1/h} = (1+ah)^{1/h} = \Big[(1+ah)^{1/(ah)}\Big]^a. \quad (5)$$

As $h \to 0$ we have $ah \to 0$, and the classical limit $\lim_{t\to0}(1+t)^{1/t} = e$ gives $(1+ah)^{1/(ah)} \to e$. Therefore the whole expression tends to $e^a = \exp\big(xf'(x)/f(x)\big)$. Taking logarithms (valid because $f > 0$ implies $\mathrm{SD}(f) > 0$) gives the more convenient identity

$$\log \mathrm{SD}(f(x)) = x\,\frac{f'(x)}{f(x)} = x\,\partial_x \log f(x). \quad (6)$$

The last equality is the chain rule: since $\partial_u \log u = 1/u$, $\partial_x \log f(x) = \frac{1}{f(x)}\cdot f'(x) = \frac{f'(x)}{f(x)}$. For a power law $f(x) = x^k$ one obtains $\mathrm{SD}(f) = e^k$: the superderivative extracts the exponent k of homogeneous growth, i.e. the number k for which $f(cx) = c^k f(x)$ for every scaling factor $c > 0$. This is the sense in which it measures relative change; the name *superderivative* is used here in the sense of multiplicative calculus [9] and is unrelated to the superderivative of supergeometry (a derivation on a superalgebra).

B. Two-dimensional extension

Definition 1 (Spatial superderivative score)**.** *Let $I : \Omega \to \mathbb{R}_{>0}$ be a positive image on a domain $\Omega \subset \mathbb{R}^2$ centred at the origin. The* superderivative score *is*

$$L(x,y) = \frac{x\,\partial_x I(x,y) + y\,\partial_y I(x,y)}{I(x,y)}, \quad (7)$$

and the 2-D superderivative is $\mathrm{SD}(I)(x,y) = \exp(L(x,y))$.

In polar coordinates, Eq. (7) becomes $L = r\,\partial_r \log I$, the radial logarithmic derivative. The score is invariant under a global intensity rescaling $I \mapsto cI$ and, after recentring, covariant under a global spatial scaling: under $I(\cdot) \mapsto I(\lambda\cdot)$ it is reparametrised as $L'(x) = L(\lambda x)$, so its value at corresponding points is unchanged.

C. From radial superderivative to isotropic log-gradient

The one-dimensional superderivative Eq. (3) [8], [9] measures relative growth along the ray emanating from the origin: the two evaluation points x and $x(1+h)$ lie on the same ray, and the quotient $f(x(1+h))/f(x)$ is the relative (multiplicative) change of f when the argument is scaled by $1+h$ along that ray. Taking the $1/h$ power and the limit converts this ratio into an instantaneous relative growth rate. In two dimensions the same idea gives the *radial superderivative*

$$\mathrm{SD}_r(I)(r,\theta) = \lim_{h\to0} \sqrt[h]{\frac{I(r\cdot(1+h),\theta)}{I(r,\theta)}} = \exp\big(r\,\partial_r \log I(r,\theta)\big). \quad (8)$$

TABLE I: Common first-derivative kernels. G_x estimates the horizontal and G_y the vertical derivative. Roberts uses the two 2×2 masks shown; the remaining operators define G_y as the transpose of G_x.

Operator	G_x / masks	Reference
Roberts	$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	[15]
Prewitt	$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$	[16]
Sobel	$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$	[17]
Frei–Chen / Isotropic	$\begin{bmatrix} -1 & 0 & 1 \\ -\sqrt{2} & 0 & \sqrt{2} \\ -1 & 0 & 1 \end{bmatrix}$	[18]
Scharr	$\begin{bmatrix} -3 & 0 & 3 \\ -10 & 0 & 10 \\ -3 & 0 & 3 \end{bmatrix}$	[19]

Because $r\,\partial_r = x\,\partial_x + y\,\partial_y$, returning to Cartesian coordinates yields exactly the spatial score

$$\log \mathrm{SD}_r(I)(x,y) = \frac{x\,\partial_x I + y\,\partial_y I}{I} = L(x,y), \tag{9}$$

i.e. Definition (7) is the direct two-dimensional transcription of the original definition [8], [9].

For edge detection the radial score has two shortcomings: it depends on the choice of origin through the weights x and y, and it measures change only in the radial direction. We therefore seek a local, origin-free and direction-independent measure of the same relative-change quantity.

The logarithmic derivative in one dimension is $\partial_x \log f = f'/f$. Its natural two-dimensional extension is the logarithmic gradient

$$\nabla \log I = \frac{\nabla I}{I}. \tag{10}$$

Taking the Euclidean norm gives

$$\|\nabla \log I\| = \frac{\|\nabla I\|}{I}, \tag{11}$$

which measures the total relative rate of change of I without privileging any direction. For a purely radial image $I = I(r)$ we have $\nabla I = (\partial_r I)\,\mathbf{e}_r$, and therefore

$$\|\nabla \log I\| = |\partial_r \log I| = \frac{|L(x,y)|}{r}. \tag{12}$$

Thus the isotropic operator equals the radial logarithmic derivative $|\partial_r \log I| = |L(x,y)|/r$ of the image intensity — not the radial superderivative itself, which is $\exp(L)$. The isotropic operator is therefore a natural local, origin-free analogue of the original radial superderivative Eq. (3), though not a derived counterpart: the superderivative is an exponential (multiplicative) object, so its exact isotropic multiplicative analogue would be $\exp(\|\nabla \log I\|)$, whereas the isotropic operator is the additive logarithmic gradient that underlies it. We keep the isotropic operator, written E_{SD} in Eq. (13), because it is the quantity that is directly comparable to the classical gradient magnitude, and we use it as an analogue motivated by the same relative-change principle rather than as a unique extension of the superderivative. In one dimension the distinction is explicit: dropping the weight x from $\log \mathrm{SD}(f) = x\,f'/f$ leaves the logarithmic derivative $\partial_x \log f = f'/f$, which is the logarithm of the *geometric* derivative $\exp(f'/f)$, not of the bigeometric one. The isotropic operator is thus a per-direction geometric log-derivative, and the superderivative contributes the relative-change principle and the radial score rather than the exact operator.

D. Why it is not a convolution

A standard first-derivative filter such as Sobel is a convolution: the same kernel is applied at every pixel. The radial score L cannot be expressed as a convolution because the weights x and y depend on the global coordinates. However, as we show below, a local and isotropic operator with the same relative-change semantics can be obtained by combining a classical derivative kernel with a pointwise division.

IV. The superderivative edge operator

A. The isotropic log-gradient edge map

Section III-C motivated the magnitude of the logarithmic gradient as the local, origin-free, isotropic analogue of the radial superderivative. We define the *relative-gradient edge-strength map* E_{SD} as

$$E_{\mathrm{SD}}(x,y) = \|\nabla \log I(x,y)\| = \frac{\sqrt{(\partial_x I)^2 + (\partial_y I)^2}}{I}. \tag{13}$$

It is the classical gradient magnitude divided pointwise by the local intensity. It should not be confused with the canonical radial score L of Eq. (7): E_{SD} is an isotropic operator inspired by the same relative-change principle, while L is the direct two-dimensional transcription of the original superderivative.

Relation to existing work: The quantity $\|\nabla \log I\|$ is not new in itself, and it is worth saying so plainly. It is the magnitude of the logarithmic (or relative) gradient, the differential form of the Weber contrast $\Delta I/I$ (the relative luminance change, named after E. H. Weber, that the eye detects against

a background I) [1], and it is what a gradient operator computes after the logarithmic transform used in homomorphic filtering [7], in logarithmic image processing [10], and in tone mapping [2]. Multiplicative (non-Newtonian) calculus, in which the geometric derivative of a positive function is $\exp(f'/f)$ and the bigeometric derivative is $\exp(xf'/f)$ [8], [26], has been developed as a self-contained theory and applied to image analysis by Florack and van Assen [11], who use the multiplicative derivative in diffusion-tensor imaging and strain analysis; they do not, however, propose the log-gradient as an edge measure. Two further families of illumination- and contrast-robust detectors are related. Phase congruency detects features where the Fourier components are maximally in phase and is invariant to image contrast [12]; the ratio-of-averages detector for synthetic-aperture radar images is designed for multiplicative speckle noise [13]. Neither is a first-derivative kernel of the image intensity: phase congruency normalises a frequency-domain local energy, and the ratio-of-averages detector uses the ratio of local means over two regions rather than a pointwise logarithmic derivative. A systematic comparison with them is left to future work. What the present paper adds is not the operator but the route to it: E_{SD} is motivated as the local, origin-free analogue of the radial superderivative [8], [9] (Eq. (7)), which fixes the two defects of that score — its dependence on the choice of origin and its blindness to tangential change — and the parametric family of Section V, which shows that the fully relative operator $p = 1$ is not the best member of its own family.

Proposition 1 (Intensity invariance)**.** *E_{SD} is invariant under a global intensity rescaling $I \mapsto cI$ for any $c > 0$.*

Proof. $\nabla \log(cI) = \nabla(\log c + \log I) = \nabla \log I$. □

Proposition 2 (Positivity)**.** *If $I > 0$ everywhere, then $E_{\mathrm{SD}} \geq 0$ everywhere.*

B. Relation to the radial score

The radial score and the log-gradient are two facets of the same idea. In polar coordinates,

$$L = r\,\partial_r \log I, \tag{14}$$

$$\|\nabla \log I\| = \sqrt{(\partial_r \log I)^2 + (\partial_\theta \log I / r)^2}. \tag{15}$$

Thus $|L|/r = |\partial_r \log I|$ is the radial component of the log-gradient, while E_{SD} is its full magnitude and therefore also captures the angular component. For images with strong radial symmetry, $|L|$ or $|L|/r$ may be preferred; for general edge detection we use E_{SD}. Figure 1 compares the canonical radial score L, its normalised form $|L|/r$, and the isotropic operator E_{SD} on two natural images, one radially structured about its centre and one not. On the retinal fundus the radial forms are adapted to the symmetry and follow the vessel arcade; on the astronaut the isotropic operator is more generally applicable. At an intensity step the canonical radial score L gives a single signed response because it is a directed derivative, whereas the magnitude forms $|L|/r$ and E_{SD} take absolute values and, through the $1/I$ (or $1/r$) weight, emphasise the dark side of the step; the result remains a single, generally asymmetric peak rather than a double-lobed or second-derivative-like profile.

C. Implementation

The operator can be evaluated in two steps. Algorithm 1 gives a concise description; any classical derivative kernel can be used in line 1.

Algorithm 1 Isotropic relative-gradient edge-strength map (implements Eq. (13), not the radial score L of Eq. (7)).

Require: Greyscale image $I > 0$, derivative kernel K
Ensure: Edge-strength map E_{SD}
1: Compute $\partial_x I = K * I$ and $\partial_y I = K^T * I$
2: $E_{\mathrm{SD}} \leftarrow \sqrt{(\partial_x I)^2 + (\partial_y I)^2}\,/\,I$
3: **return** E_{SD}

Step 1 is a convolution; step 2 is pointwise. The whole operator therefore has the same computational complexity as a classical gradient magnitude plus one division per pixel. The two-step procedure estimates Eq. (13) with the same finite-difference kernel as a classical gradient magnitude: the kernel provides an estimate of ∇I, and the pointwise division by I yields $\|\nabla I\|/I$. The only approximation inherited from classical methods is the choice of finite-difference kernel.

The two factors of Eq. (13) can also be discretised in the opposite order. Since $\|\nabla I\|/I = \|\nabla \log I\|$ in the continuum, one may instead apply the kernel to the logarithmic image and take $\|K * \log I\|$. The two estimates agree away from a step but differ at one: the first weights the intensity difference by the value at a single pixel, so at a step from I_1 to I_2 it gives $|I_2 - I_1|/I_1$ on one side and $|I_2 - I_1|/I_2$ on the other, whereas the second gives the symmetric logarithmic difference $|\log(I_2/I_1)|$ on both sides. The two-step form is the default because it reuses the classical kernels directly; Section VI-L compares the two.

A practical caveat is that real images may contain zero-valued pixels. A small offset ε can be added to the denominator, giving $E_\varepsilon = \|\nabla I\|/(I + \varepsilon)$. This breaks exact intensity invariance in the darkest regions but avoids numerical instability.

V. A parametric family

The absolute gradient and the log-gradient are two extreme points of a single parametric family:

$$E_p(x, y) = \frac{\|\nabla I(x, y)\|}{I(x, y)^p}, \qquad p \in [0, \infty). \tag{16}$$

For $p = 0$ we recover the classical absolute gradient magnitude. For $p = 1$ we recover the superderivative/log-gradient magnitude. Values $p \in (0, 1)$ give an intermediate behaviour: the response is partially normalised by the local intensity without becoming as aggressive as the full log-gradient in dark regions.

Proposition 3 (Intensity invariance of the family)**.** *E_p is invariant under $I \mapsto cI$ if and only if $p = 1$.*

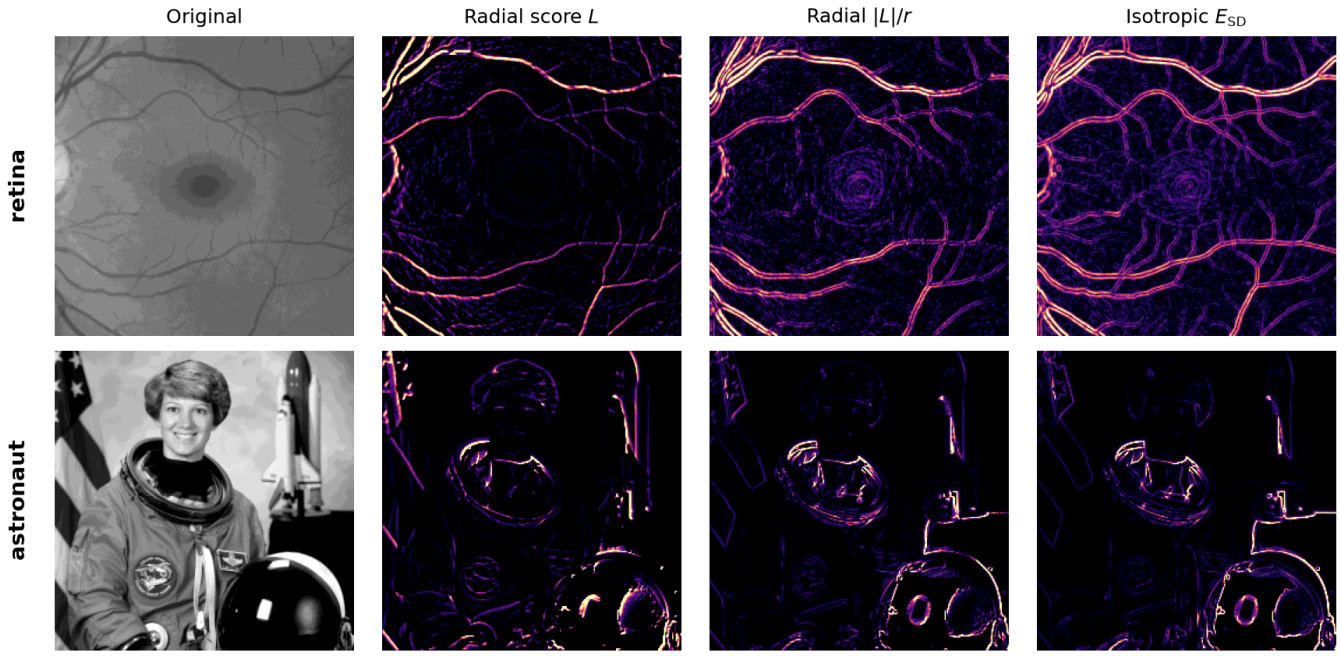


Fig. 1: Comparison of the canonical radial score $L = (xI_x + yI_y)/I$, its normalised version $|L|/r$, and the isotropic superderivative $E_{\mathrm{SD}} = \|\nabla I\|/I$ (Eq. (13)). The retinal fundus is radially structured about the image centre (radial illumination falloff, macula at the centre, vessel arcade sweeping around it), which is the regime the radial forms assume; the astronaut has no such symmetry, and the isotropic form is the recommended operator for general edge detection.

Proof. $\|\nabla(cI)\| = c\|\nabla I\|$ and $(cI)^p = c^p I^p$. Therefore $E_p(cI) = c^{1-p}\|\nabla I\|/I^p = c^{1-p}E_p(I)$, which equals $E_p(I)$ if and only if $p = 1$. □

Remark 1 (Gamma transforms)**.** *Since* $\nabla \log(I^\gamma) = \gamma \nabla \log I$*, a gamma transform multiplies* E_{SD} *by the constant* γ *and leaves the normalised edge map unchanged, while for* $p \neq 1$ *it gives the position-dependent factor* $\gamma\, I^{(\gamma-1)(1-p)}$*. The* $p = 1$ *operator is thus covariant under gamma, not merely scale-invariant — one more property pulling against the experiments, which prefer a small interior exponent (*$p = 0.25$*).*

Thus the full log-gradient ($p = 1$) is the unique intensity-invariant member of the family; members with $p < 1$ trade some invariance for numerical stability.

The above observation is important in practice: exact intensity invariance requires $p = 1$, but the BSDS500 experiments below show that the best empirical performance is obtained with a small interior exponent ($p = 0.25$). The slight loss of invariance is compensated by better stability in dark regions.

A. Regularisation and robust alternatives

The raw log-gradient can become numerically unstable or visually faint when I is small or when the relative change is very small. Several regularised variants keep the same relative-change semantics while improving robustness:

- **Additive regularisation:** $E_{\alpha,p} = \|\nabla I\|/(I^p + \alpha)$ with $\alpha > 0$. In bright regions this behaves like E_p; in dark regions it behaves like an absolute gradient scaled by $1/\alpha$, which avoids division-by-zero and prevents the map from collapsing to black.
- **Local-mean denominator:** replace I^p by $(\mu_\rho * I)^p$, where the locally averaged intensity $\mu_\rho * I$ is computed with a box/Gaussian filter of radius ρ and then raised to the same exponent. This makes the denominator robust to shot noise while retaining the relative-change interpretation at scales larger than ρ.

Figure 2 compares these variants on natural images.

VI. Experiments

A. Setup

We generated five synthetic 256×256 images with known ground-truth edges:

- **Circles:** concentric annuli with five boundaries.
- **Square:** a bright square on a dark background.
- **Checkerboard:** an 8×8 checkerboard.
- **Triangle:** a bright triangle on a dark background.
- **Ramp:** a linear intensity ramp with an additive step.

The annulus boundaries, the triangle boundary, and the square boundary are computed analytically from the generating mask, so the ground truth does not depend on any detector; the checkerboard ground truth is the grid of internal boundaries of the generating mask, and the ramp ground truth is the column of the additive step. For each image we computed the edge-strength maps of the proposed operator, the radial score $|L|/r$, and the classical operators Sobel, Scharr, Prewitt, Roberts and Canny. The proposed and radial operators use the

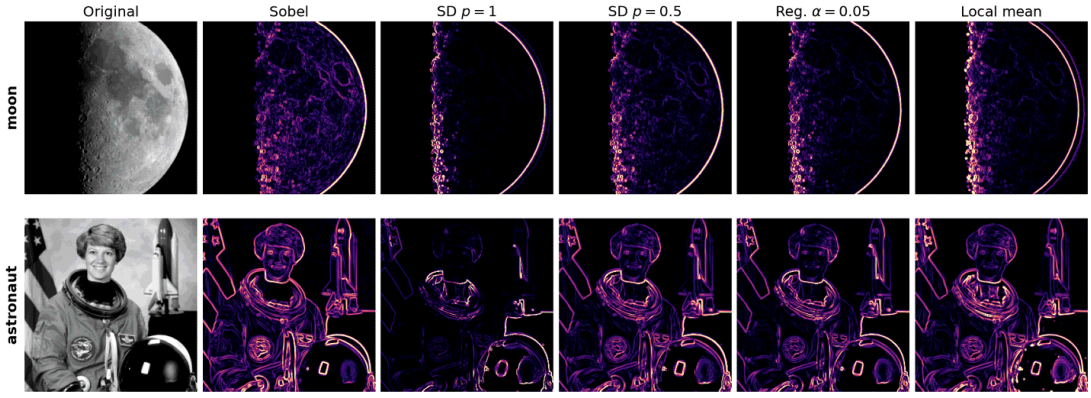


Fig. 2: Alternative superderivative formulations, one row per image and one column per formulation: original, Sobel, the log-gradient $p = 1$, the parametric $p = 0.5$, the additively regularised $\alpha = 0.05$ and the local-mean denominator. The lunar image is included because its dark sky is where an unregularised division by I is worst behaved, which is what the regularised and local-mean variants are for. Maps are scaled to their 99th percentile, so a single saturated pixel in the dark sky cannot blacken the rest of the panel.

Sobel kernel throughout, while each classical operator uses its own kernel, so that the parametric comparison is an ablation against the same kernel for the proposed family; the input is clipped to $[10^{-3}, 1]$, each map is scaled to $[0, 1]$ by its minimum and maximum, thresholded at 23 levels linearly spaced in $[0.02, 0.90]$, and thinned to a one-pixel skeleton (scikit-image `skeletonize`) before matching. The best F1 over the threshold sweep was computed with a one-pixel localisation tolerance, as in the BSDS benchmark (Section VI-F). Here $F_1 = 2PR/(P+R)$ is the harmonic mean of precision P and recall R at a given threshold, and the "best" is taken over the threshold sweep; a one-pixel localisation tolerance credits a predicted edge that falls within one pixel of the ground truth, instead of requiring exact coincidence. A strict pixel-exact rule is not used to rank detectors, because it charges a one-pixel offset as both a false positive and a false negative.

B. Quantitative results

Table II reports F1 scores. On these piecewise-constant images most first-derivative methods, including the proposed one, recover the edges almost perfectly. The main exceptions are Roberts on the triangle, whose 2×2 response misses part of the boundary (0.871), and the radial score, which fails on the ramp, whose gradual slope is tangential to the radial direction over most of the image, and loses on the checkerboard, whose axis-aligned internal boundaries are mostly perpendicular to the radial direction away from the centre. A strict pixel-exact rule would charge the one-pixel offset of detectors such as Roberts, and of Canny on the ramp, as both a false positive and a false negative, an artefact of the metric rather than a property of the operators; we therefore report only the tolerance-based rule throughout. Because these piecewise-constant images cannot separate the operators, Section VI-K isolates the regime where the relative and absolute weights must differ.

C. Visual comparison

Figure 3 shows the edge-strength maps. The superderivative operator gives thin, clean responses at intensity discontinuities, similar to Sobel and Scharr. Because it normalises by the local intensity, its response is more uniform across objects of different brightness.

D. Intensity-scale invariance

To verify the main theoretical property experimentally, we took a synthetic piecewise-constant image and multiplied it by $c \in \{0.3, 1.0, 3.0\}$. The Sobel edge-strength map scales linearly with c ($\|\nabla(cI)\| = c\|\nabla I\|$), whereas the superderivative map remains unchanged because the factor c cancels in $\|\nabla(cI)\|/(cI)$. Figure 4 shows the maps on a common absolute scale: the Sobel maps become brighter as c increases, while the superderivative maps are visually identical.

E. Natural images

Figure 5 shows edge-strength maps for six standard natural images. The superderivative operator produces responses comparable to Sobel and Scharr, with slightly cleaner behaviour in regions of very different brightness because of the intensity normalisation.

F. BSDS500 benchmark

We evaluated the detectors on the 100 images of the BSDS500 validation split with the official boundary benchmark of Arbeláez *et al.* [28]. The matching code is the original Berkeley C++ implementation (the same `correspondPixels` used by the released benchmark), so the pixel correspondence is the optimal, minimum-cost assignment of the Berkeley CSA solver rather than an approximation. Following the official defaults, each of the five (or six) human boundary maps is used separately, the prediction is thresholded at the 99 levels $\{0.01, \ldots, 0.99\}$ and thinned to a one-pixel

TABLE II: Best F1 on synthetic ground-truth edges under a threshold sweep with one-pixel localisation tolerance (Section VI-F). Highest score per image is in **bold**. Most first-derivative methods are within about 0.02 of each other; the exceptions are Roberts on the triangle and the radial score on the ramp and checkerboard, where their respective definitions are a genuine handicap.

Image	$E_{0.25}$ (Sobel)	$\lvert L\rvert/r$	Sobel	Scharr	Prewitt	Roberts	Canny
Circles	0.986	0.985	0.986	**0.988**	0.984	0.979	0.980
Square	**1.000**	**1.000**	0.998	0.998	0.998	0.998	0.996
Checkerboard	**1.000**	0.942	0.998	0.998	0.998	**1.000**	0.998
Triangle	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.871	0.998
Ramp	0.996	0.188	0.996	0.996	0.996	**1.000**	0.996

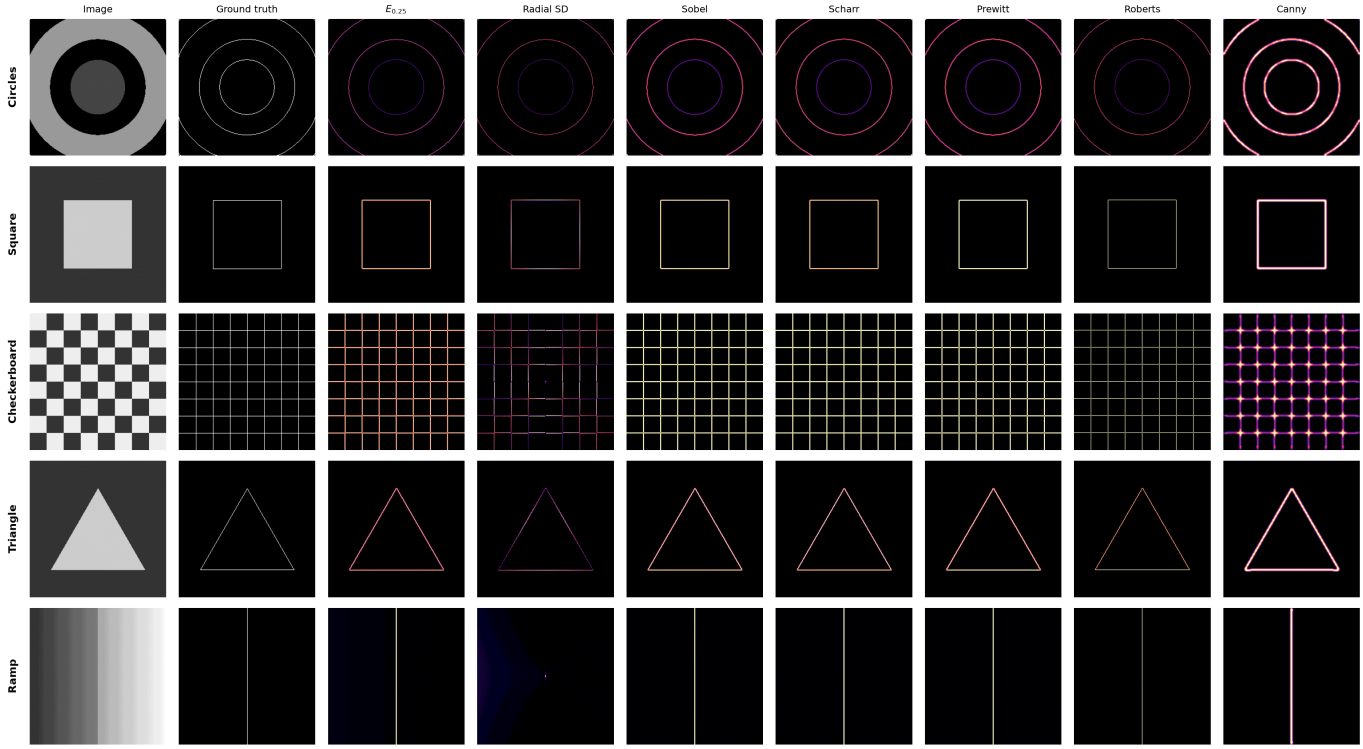


Fig. 3: Edge-strength maps for synthetic images. Rows correspond to concentric annuli, square, checkerboard, triangle and intensity ramp; columns show the original image, ground truth, and the outputs of the superderivative operator ($p = 0.25$), the radial score, Sobel, Scharr, Prewitt, Roberts and Canny. The radial score on the ramp collapses to a single spot because the gradual slope contributes only a weak, coordinate-weighted response, while the additive step produces a strong localised peak that dominates the normalised colour scale.

skeleton (scikit-image `skeletonize`), and the thresholded map is matched to each human map within 0.0075 of the image diagonal with an outlier cost of 100. We report the optimal dataset scale (ODS, the best F-measure obtained with a single threshold shared by all images), the optimal image scale (OIS, the best F-measure obtained with the threshold tuned for each image) and the average precision (AP). ODS and OIS take the maximum over the fixed 99-threshold grid rather than interpolating between thresholds, and AP follows the official recall interpolation. The detectors are our own implementations (finite-difference kernels and the Canny of SCIKIT-IMAGE, with the automatic thresholds of that library and tuned per image over $\sigma \in \{0.5, 1, 1.5, 2, 2.5\}$), and the input is clipped to $[10^{-3}, 1]$ to keep the relative operators finite in dark regions. Each edge map is scaled to $[0, 1]$ before thresholding, by its minimum and maximum in the main table and by its 99th percentile in the robustness check of Section VII.

Table III reports the result. The tuned Canny is the best method in ODS and OIS (0.679 and 0.674), and its ODS margin over the best member of the parametric family is large (+0.088) and significant on a paired t-test over the per-image OIS (+0.031, $p \approx 10^{-6}$). Canny's σ is tuned per image here, but its advantage does not come from that tuning: with a single global σ chosen by ODS the same detector still reaches ODS 0.658 and OIS 0.653, far ahead of the unsmoothed family, so the comparison is not an oracle. The Canny maps of SCIKIT-IMAGE are binary, so their threshold sweep is degenerate and their ODS (pooled counts) can exceed their OIS (mean per-image best), because the two aggregate the per-image F-measures differently; the ordering against the continuous maps is unaffected. Among the first-derivative kernels the parametric family peaks at $p = 0.25$ on the validation split, where it has the best ODS (0.591) and AP (0.651) of the

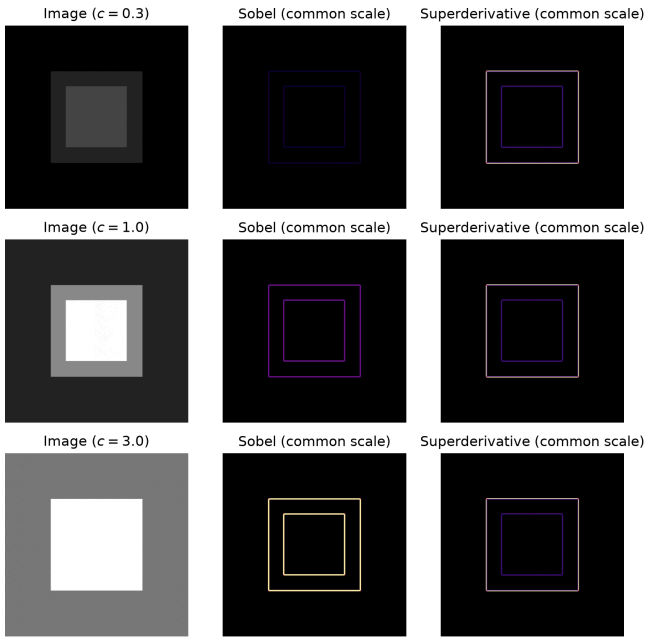


Fig. 4: Intensity-scale invariance. Rows correspond to global scalings $c = 0.3, 1.0, 3.0$ of the same image. Both edge maps are displayed on a common absolute scale, without per-image normalisation. Sobel's absolute response grows linearly with c; the superderivative absolute response remains unchanged (up to numerical round-off). The displayed image for $c = 3$ is saturated, but the operators process the unclipped values.

family and beats the raw Sobel, Prewitt and Scharr kernels by 0.004–0.009 in ODS; its per-image OIS advantage is small but significant ($+0.005$ to $+0.006$, bootstrap 95% CI $[+0.002, +0.009]$, $p < 0.01$), and $p = 0.5$ is statistically indistinguishable from $p = 0.25$ in OIS (-0.0001, 95% CI $[-0.004, +0.005]$, $p = 0.96$) but has a lower ODS (0.538). Table VI repeats the comparison on the 200 held-out test images, which were not used to choose p. The per-image OIS and AP advantages replicate ($+0.004$ to $+0.006$ in OIS, bootstrap 95% CI $[+0.002, +0.008]$, $p < 10^{-4}$; $+0.003$ to $+0.004$ in AP), but the single-threshold ODS margin over Sobel and Prewitt does not: on the held-out split $p = 0.25$ (0.588) is statistically tied with Sobel (0.589) and marginally below Prewitt (0.591), while still above Scharr (0.586). The interior optimum is therefore robust in the per-image metric, but the headline ODS gain over the kernels is confined to the validation split; we state both and do not claim the ODS margin as a held-out result. The full superderivative $p = 1$ is *worse* than the classical kernels under min-max, by 0.13 in ODS and 0.046 in OIS (bootstrap 95% CI $[-0.073, -0.021]$, $p \approx 10^{-3}$), because dividing by the local intensity amplifies noise in dark regions, and $p = 1.5$ is worse still. The interior optimum is robust to the normalisation. Table V repeats the benchmark with a robust percentile normalisation: the family still peaks at a small interior exponent ($p = 0.5$, ODS 0.617, with $p = 0.75$ and $p = 0.25$ at 0.613 and 0.611) and still beats the kernels (0.600–0.601), so the optimum is not an artefact of the min-max threshold grid. The collapse of the fully relative endpoint, however, is: $p = 1$ is below the kernels under min-max but statistically indistinguishable from them under percentile (per-image OIS difference from Sobel -0.002, 95% CI $[-0.015, +0.010]$, $p = 0.73$), so its ranking is a threshold-grid effect of the min-max scaling rather than a property of the relative gradient. The two discretisations are likewise comparable under percentile (log-image Sobel 0.604 against two-step Sobel 0.601). Two baselines that the construction invites confirm the picture. Discretising the log-gradient on the logarithmic image, $\|K * \log I\|$, gives ODS 0.578 and AP 0.626, close to the two-step Sobel kernel, so the two-step form is not disadvantaged by its discretisation. The additively regularised variant ($p = 0.5$, $\alpha = 0.05$) recovers ODS 0.574 with OIS 0.644 and AP 0.649, above the full $p = 1$ and competitive with the classical kernels, which is what the regularised denominator is for. Finally, the comparison with Canny is not like-for-like: Canny adds non-maximum suppression and hysteresis to Gaussian smoothing, whereas the family here is only a thresholded edge-strength map. To separate the effect of smoothing from that of the full pipeline, we apply the same Gaussian pre-smoothing to the classical kernels and to the family, with a single σ chosen by ODS for each (as for the single-σ Canny above). This changes the picture: $E_{0.25}$ at $\sigma = 2.0$ reaches ODS 0.659, OIS 0.684 and AP 0.713, against ODS 0.658, OIS 0.653 and AP 0.652 for the single-σ Canny and ODS 0.586, OIS 0.637 and AP 0.647 for unsmoothed Sobel. The smoothed family therefore matches the single-σ Canny in ODS and leads it in OIS and AP, while the tuned Canny remains ahead in ODS. Smoothed Sobel at $\sigma = 2.0$ (ODS 0.650, OIS 0.679, AP 0.707) shows that most of the gain over the unsmoothed kernels comes from the smoothing rather than the normalisation, the latter adding a small margin on top. This does not isolate the contribution of non-maximum suppression and hysteresis, which the family does not implement. The value of the construction is therefore the family rather than its endpoint: a small interior intensity normalisation helps by a small but detectable margin, the optimum is interior, and a tuned Canny remains ahead of the unsmoothed kernels.

Figure 6 summarises the results under min-max. The parametric family peaks at a small interior exponent under both normalisations: $p = 0.25$ under min-max and $p = 0.5$ under percentile, with $p = 0.75$ and $p = 0.25$ close behind. The peak is shallow in OIS, and the drop of the large exponents under min-max is the statistically solid effect.

G. Outlook: mathematical alternatives and extensions

The framework invites several refinements without introducing post-processing; none of them is evaluated on BSDS500 here, and they are listed as outlook rather than results.

Parametric exponent. The family $E_p = \|\nabla I\|/I^p$ lets the user choose the degree of intensity normalisation. The BSDS500 experiments suggest $p \in [0.25, 0.5]$ as a good compromise between invariance and stability.

Regularised denominator. Replacing I by $I + \varepsilon$ avoids overflow in dark regions. The resulting map $E_\varepsilon = \|\nabla I\|/(I + \varepsilon)$ tends to the log-gradient when $\varepsilon \ll I$ and to a scaled

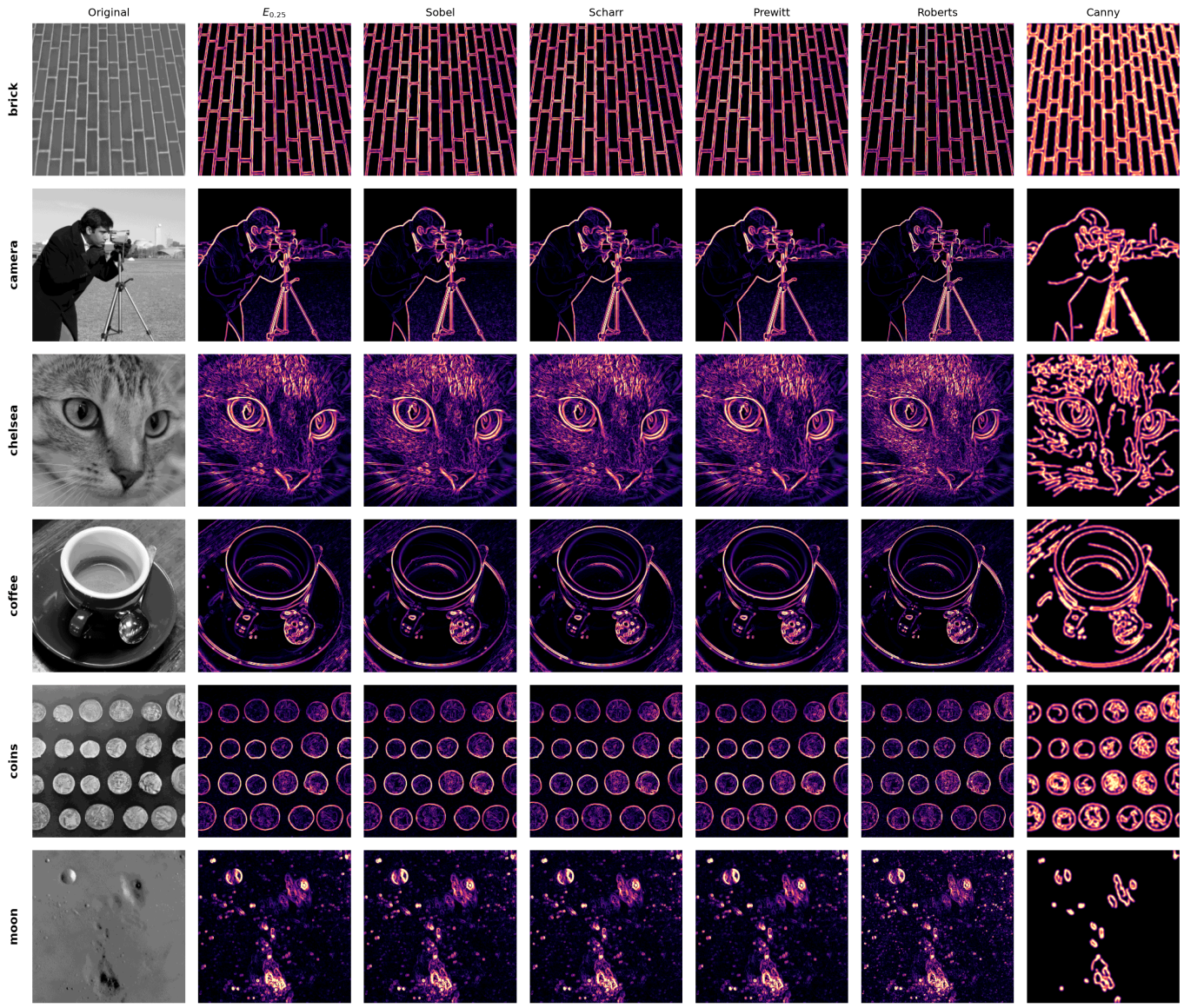


Fig. 5: Edge-strength maps for natural images (brick, camera, chelsea, coffee, coins, moon). Columns show the original image and the outputs of the superderivative operator ($p = 0.25$), Sobel, Scharr, Prewitt, Roberts and Canny.

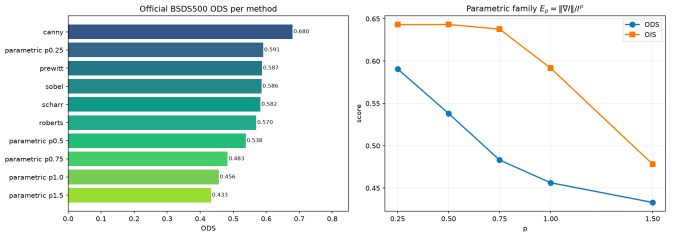


Fig. 6: Official BSDS500 results on the validation split. Left: ODS for each method. Right: ODS and OIS as a function of the parametric exponent p.

absolute gradient, $\|\nabla I\|/\varepsilon$, when $\varepsilon \gg I$; it is a one-parameter family distinct from E_p, and the two can be combined as $\|\nabla I\|/(I^p + \alpha)$. The additively regularised variant is included in the BSDS500 comparison above.

Radial score for symmetric objects. For images with approximate radial symmetry, the score $L = r\,\partial_r \log I$ or its normalised version $|L|/r$ can be used directly. It measures relative change under a zoom and is therefore naturally scale-covariant after recentring: under a spatial rescaling it is reparametrised, not multiplied by a factor, so its value at corresponding points is unchanged.

Higher-order operators. Just as the Laplacian extends the gradient, one can take the divergence of the relative-gradient field or apply the multiplicative derivative twice, giving a relative-curvature operator. A systematic treatment is left for future work; the expression $\partial_x(x\partial_x \log I) + \partial_y(y\partial_y \log I)$ used here is only a candidate and is not the second bigeometric derivative.

Scale-space. All kernels in Table I can be applied after Gaussian smoothing at scale σ. A multi-scale superderivative

TABLE III: Official BSDS500 boundary benchmark on the 100 validation images (ODS/OIS/AP; five human maps used separately; 99 thresholds; thresholded prediction thinned to one pixel; optimal-assignment matching within 0.0075 of the image diagonal; outlier cost 100; edge maps scaled to $[0, 1]$ by their minimum and maximum). Our own detector implementations, so the values are not directly comparable to published results for other detectors. Higher is better; best in **bold**.

Method	ODS	OIS	AP
Canny (tuned σ)	**0.679**	**0.674**	0.613
Canny (single σ)	0.658	0.653	**0.652**
Parametric $p = 0.25$	0.591	0.643	0.651
Prewitt	0.587	0.638	0.647
Sobel	0.586	0.637	0.647
Scharr	0.582	0.637	0.645
Log-image Sobel	0.578	0.633	0.626
Regularised $p = 0.5$	0.574	0.644	0.649
Roberts	0.570	0.629	0.631
Parametric $p = 0.5$	0.538	0.643	0.647
Parametric $p = 0.75$	0.483	0.638	0.641
Superderivative $p = 1$	0.456	0.592	0.644
Parametric $p = 1.5$	0.433	0.478	0.644

TABLE IV: Effect of Gaussian pre-smoothing on the same BSDS500 validation split (min-max normalisation). A single σ is chosen by ODS for each method, as for the single-σ Canny. The unsmoothed kernels are compared against Canny, but once both are smoothed the family matches the single-σ Canny in ODS and leads it in OIS and AP, while the tuned Canny remains ahead in ODS. Higher is better; best in **bold**.

Method	ODS	OIS	AP
$E_{0.25}$, $\sigma = 2.0$	**0.659**	**0.684**	**0.713**
Canny (single σ)	0.658	0.653	0.652
Sobel, $\sigma = 2.0$	0.650	0.679	0.707
$E_{0.25}$, $\sigma = 1.0$	0.631	0.667	0.686
Sobel, $\sigma = 1.0$	0.620	0.662	0.680
$E_{0.25}$, $\sigma = 0.5$	0.609	0.650	0.662
Sobel, $\sigma = 0.5$	0.593	0.644	0.657
$E_{0.25}$ (no smoothing)	0.591	0.643	0.651
Sobel (no smoothing)	0.586	0.637	0.647

TABLE V: Same official BSDS500 benchmark as Table III but with each edge map scaled to $[0, 1]$ by its 99th percentile instead of its minimum and maximum. The interior optimum of the family and its margin over the kernels survive the change; the min-max collapse of the larger exponents does not. Higher is better; best in **bold**.

Method	ODS	OIS	AP
Canny (tuned σ)	**0.679**	**0.673**	0.611
Canny (single σ)	0.653	0.627	0.619
Parametric $p = 0.5$	0.617	0.648	0.631
Regularised $p = 0.5$	0.617	0.648	0.632
Parametric $p = 0.75$	0.613	0.645	0.629
Parametric $p = 0.25$	0.611	0.647	**0.638**
Log-image Sobel	0.604	0.637	0.614
Parametric $p = 1$	0.601	0.639	0.620
Sobel	0.601	0.641	0.633
Prewitt	0.601	0.643	0.633
Scharr	0.600	0.641	0.632
Roberts	0.595	0.634	0.617
Parametric $p = 1.5$	0.573	0.620	0.599

operator would compute E_p at several scales and combine the responses.

H. Parametric family visualisation

Figure 7 shows the parametric edge-strength maps $E_p = \|\nabla I\|/I^p$ for several values of p. As p increases, the response becomes more uniform across regions of different brightness, but the division by smaller intensities also amplifies noise in dark areas.

I. Noise robustness

Figure 8 compares the detectors as additive Gaussian noise increases. Because the relative operator divides by the local intensity, additive noise is amplified where I is small, so its advantage over the classical kernels at low noise reverses at high noise: measured as best F1 against the synthetic ground truth of Fig. 3, at $\sigma = 0.02$ the relative $p = 0.5$ map is marginally ahead of Sobel (1.000 vs. 0.998), at $\sigma = 0.05$ they are comparable (1.000 vs. 0.998), and at $\sigma = 0.10$ the relative map collapses (0.265 for $p = 0.5$, 0.038 for $p = 1$) while Sobel holds at 0.992. This is the same dark-region amplification that limits $p = 1$ on BSDS500, and it is what the regularised denominator of Section V-A is for; the low-light experiment below makes the point in the spatial domain.

J. Low-light and texture

Figures 9 and 10 show two additional regimes: images under simulated low-light conditions and highly textured natural surfaces. Under simulated low light the proposed operator degrades less than the classical kernel in the map-difference sense: the mean absolute difference between the normal-light and low-light maps, relative to the normal-light mean, is 0.249 at $p = 0.5$ and 0.235 for the regularised variant, against 0.321 for Sobel, while the fully relative $p = 1$ is the worst at 0.603. This measure is a self-consistency check, not an accuracy metric, and low light here is a gamma compression plus additive noise rather than the global rescaling of the invariance proposition. On textured images the operator likewise suppresses weak texture gradients through the same normalisation, at the cost of some faint true edges.

K. Controlled checks of relative versus absolute change

Because the global rescaling of the invariance proposition cancels under any per-image normalisation, the invariance is only observable without normalisation (Fig. 4). Its practical content is that equal *contrast ratios* $\Delta I/I$ receive equal response. Figure 11 and the numbers below isolate that content on two analytic images. On two steps of the same absolute size 0.05 but on backgrounds 0.10 and 0.50, the ratio of the bright-step to dark-step peak response is 1.000 for Sobel and 0.273 for E_{SD}; on two steps of the same contrast ratio 0.20 but on the same two backgrounds, it is 5.000 for Sobel and 1.000 for E_{SD}. On a reflectance image with faint edges, $I = R{\cdot}L$, multiplied by a smooth illumination field L, the best F1 against the edges of R is 1.000 for E_{SD} ($p = 1$) against 0.998 for Sobel, and E_{SD} is already at the level reached by Sobel after exact division by L. The gain is small for a smooth field, as expected, and grows with the dynamic range of the illumination; it is this contrast equalisation, not the global rescaling of Fig. 4, that the relative normalisation buys.

TABLE VI: Same protocol as Table III on the 100 validation and the 200 held-out test images (min-max normalisation). The exponent $p = 0.25$ was chosen on validation only. The per-image OIS and AP advantages over the kernels replicate on the held-out split; the ODS margin over Sobel and Prewitt does not. Our own detector implementations; higher is better; best in **bold**.

	Validation (100)			Test (200)		
Method	ODS	OIS	AP	ODS	OIS	AP
Canny (tuned σ)	**0.679**	**0.674**	0.613	**0.681**	**0.674**	0.617
Parametric $p = 0.25$	0.591	0.643	**0.651**	0.588	0.637	**0.646**
Prewitt	0.587	0.638	0.647	0.591	0.633	0.642
Sobel	0.586	0.637	0.647	0.589	0.633	0.642
Scharr	0.582	0.637	0.645	0.586	0.631	0.641
Superderivative $p = 1$	0.456	0.592	0.644	0.458	0.599	0.645

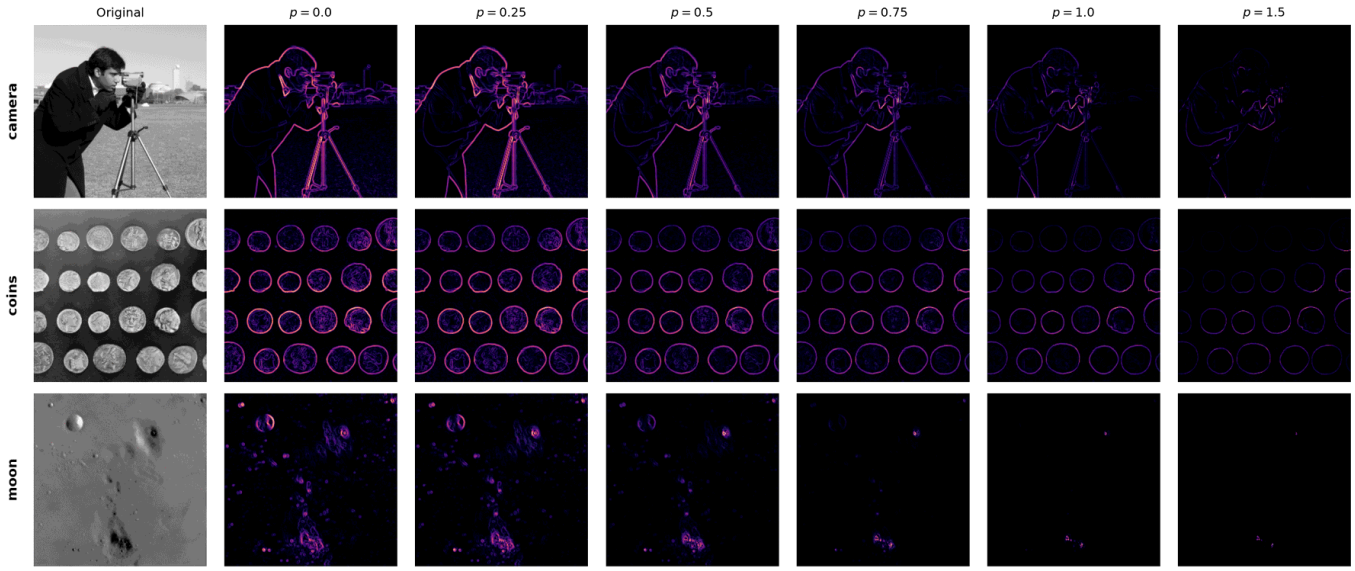


Fig. 7: Parametric family $E_p = \|\nabla I\|/I^p$ for $p = 0, 0.25, 0.5, 0.75, 1, 1.5$ on natural images. $p = 0$ is the absolute gradient; $p = 1$ is the superderivative/log-gradient.

L. Discretisation of the relative gradient

The two discretisations of $\|\nabla \log I\|$ agree away from a step but not at one. On the synthetic piecewise-constant images of Fig. 3, the two-step form $\|K * I\|/I$ reaches best F1 1.000 and the log-image form $\|K * \log I\|$ reaches 0.998: the two are within the tolerance of the benchmark, so the two-step form used throughout is a sound discretisation and the small difference is not what drives the $p = 1$ behaviour on BSDS500, which is a dark-region effect rather than a discretisation artefact.

VII. Discussion

On the official BSDS500 validation split a small interior exponent ($p \approx 0.25$–0.5) gives a small but statistically detectable improvement over the classical first-derivative kernels under both a min-max and a robust percentile normalisation of the edge maps, so the interior optimum is not an artefact of the threshold grid. On the held-out test split the per-image OIS and AP advantages replicate, while the single-threshold ODS margin over Sobel and Prewitt does not (Table VI), so the positive claim is stated per-image and the ODS margin is confined to the validation split. The fully relative operator ($p = 1$) is never the best member and its ranking is normalisation-sensitive: division by a small I amplifies dark-region noise, and the min-max scaling of a heavy-tailed map turns that noise into a threshold-grid effect that the percentile normalisation removes. On the synthetic piecewise-constant geometry most first-derivative methods recover the edges almost equally well, with Roberts on the triangle as the exception. The conceptual advantage of the framework—making the absolute-to-relative passage explicit—and the verified intensity-scale invariance at $p = 1$ (Fig. 4) remain, but they do not by themselves select the empirically best member of the family.

Its main limitation is the division by I. In very dark regions a small amount of noise is amplified because the denominator is close to zero. In practice this is mitigated by adding a small offset ε to the denominator, at the cost of a slight loss of exact invariance in the darkest areas. A second limitation is that the full radial score L is not shift-invariant, because the weights x and y are measured from the image centre. The isotropic version E_{SD} restores shift invariance and is the recommended form for general edge detection. A third, methodological limitation is that neither the unsmoothed nor the merely smoothed first-derivative kernels are like-for-like

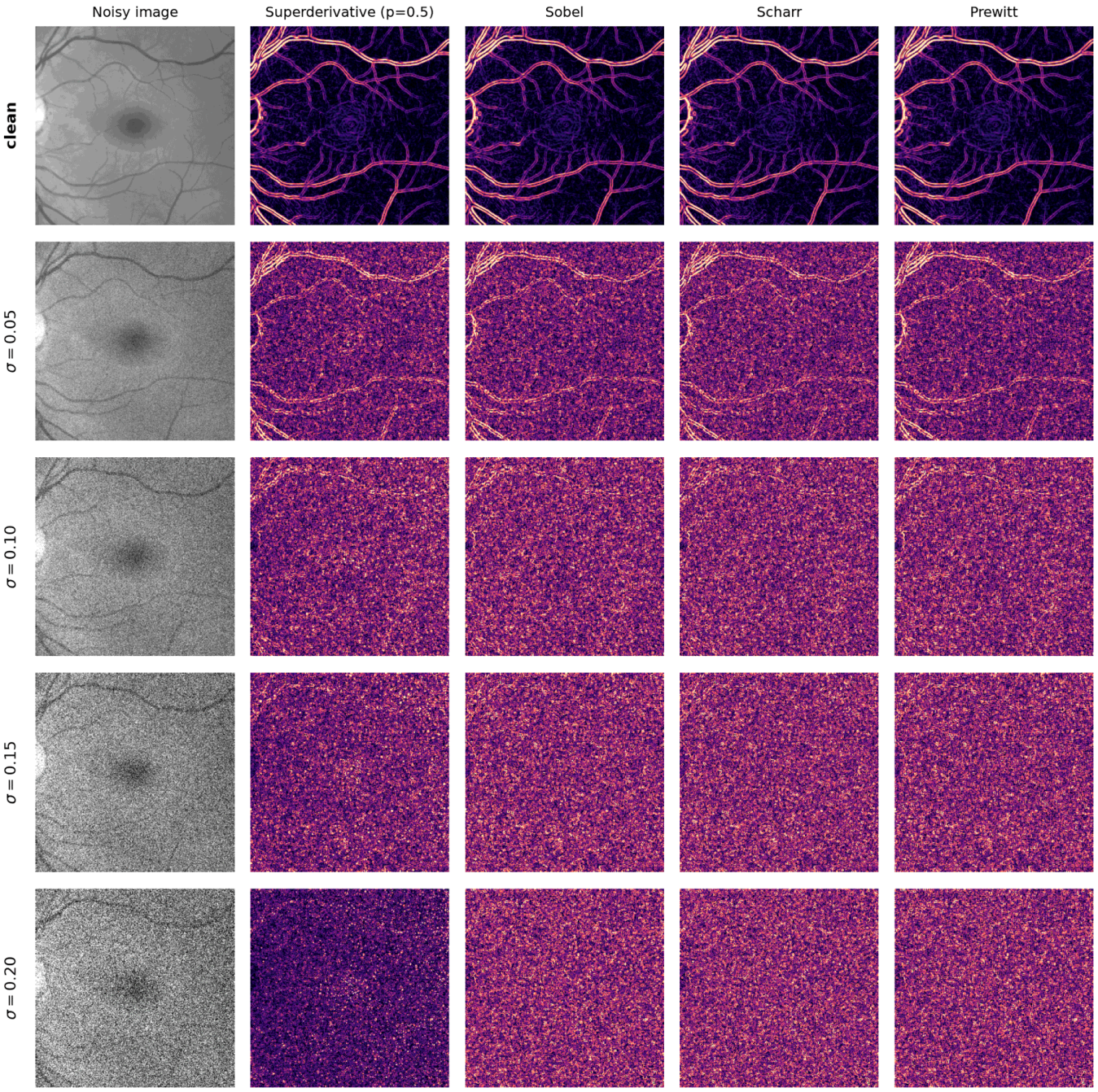


Fig. 8: Robustness to additive Gaussian noise. Rows correspond to increasing noise standard deviation; columns show the noisy input and the outputs of the superderivative ($p = 0.5$), Sobel, Scharr and Prewitt operators.

comparisons with Canny, which also applies non-maximum suppression and hysteresis. Table IV shows that once the same Gaussian smoothing is applied, the family matches the single-σ Canny in ODS and leads it in OIS and AP, but the tuned Canny remains ahead in ODS; the comparison therefore separates the effect of smoothing from that of the full Canny pipeline only partially.

Future work includes adding non-maximum suppression and hysteresis thresholding (as in Canny), mathematical robustifications of the general formulation and its discretisation, multi-scale extensions, and applying the operator to image enhancement, segmentation and feature detection.

VIII. Conclusions

We have revisited the log-gradient edge operator and motivated it by analogy with the superderivative, a multiplicative derivative in which the independent variable is perturbed multiplicatively. The operator measures relative rather than

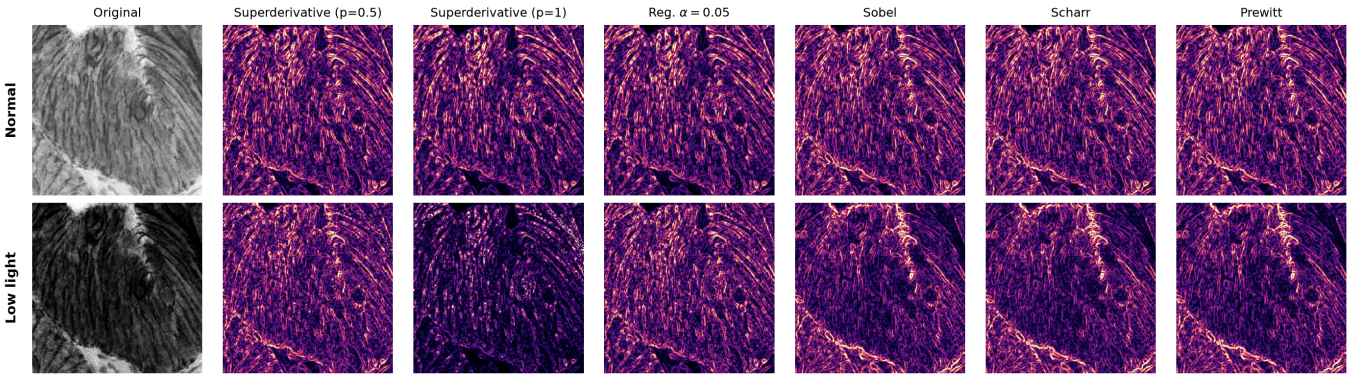


Fig. 9: Edge detection under normal (top row) and simulated low-light (bottom row) conditions. Low light is a gamma compression $I^{2.5}$ followed by additive sensor noise $\sigma = 0.02$, *not* the global rescaling $I \mapsto cI$ of the invariance proposition, under which the maps are identical to machine precision (Fig. 4). What degrades the response is the noise, not the loss of light. Figures are means over ten noise realisations with a fixed seed (42); the spread is ± 0.002 for $p = 0.5$ and the regularised variant and ± 0.021 for $p = 1$.

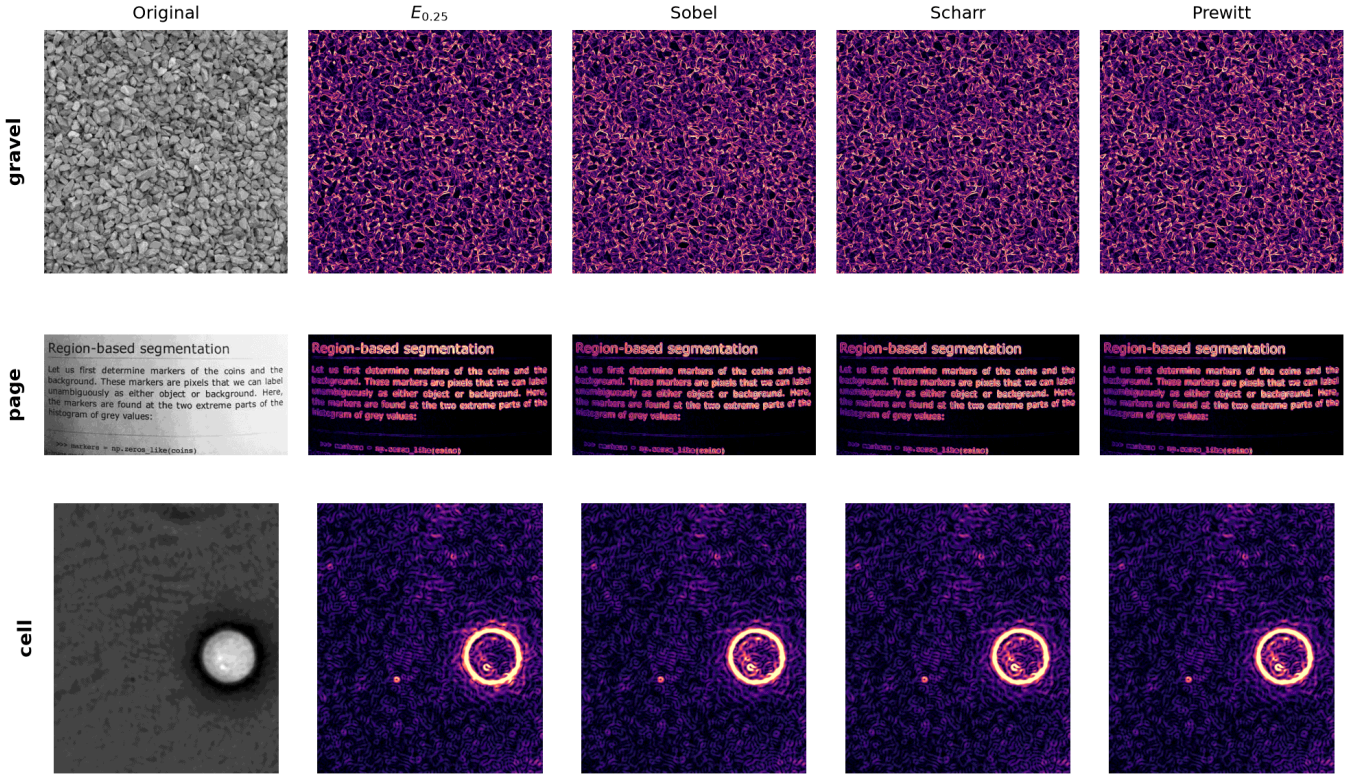


Fig. 10: Edge detection on textured images (gravel, page, cell). The superderivative column uses $p = 0.25$.

absolute intensity change, is invariant under global intensity rescaling, and can be implemented in two simple steps. A parametric family interpolates between the absolute gradient and the log-gradient; on synthetic images most first-derivative methods recover the edges almost perfectly, with Roberts on the triangle as the exception, while on the official BSDS500 benchmark the best members are small intermediate exponents ($p \approx 0.25$–0.5), edging the classical kernels by a small but detectable margin in the per-image metric; the single-threshold ODS margin over the kernels holds on the validation split but not on the held-out test split, and the fully relative endpoint $p = 1$ is never the best member, its ranking depending on the edge-map normalisation. With the same Gaussian smoothing that the single-σ Canny uses, the smoothed family matches it in ODS and leads it in OIS and AP, while the tuned Canny remains ahead in ODS; the unsmoothed comparison therefore understates the smoothed family, though it does not isolate the effect of non-maximum suppression and hysteresis. The contribution is therefore the construction, the invariance it buys, and a clear account of when it helps — a small gain at

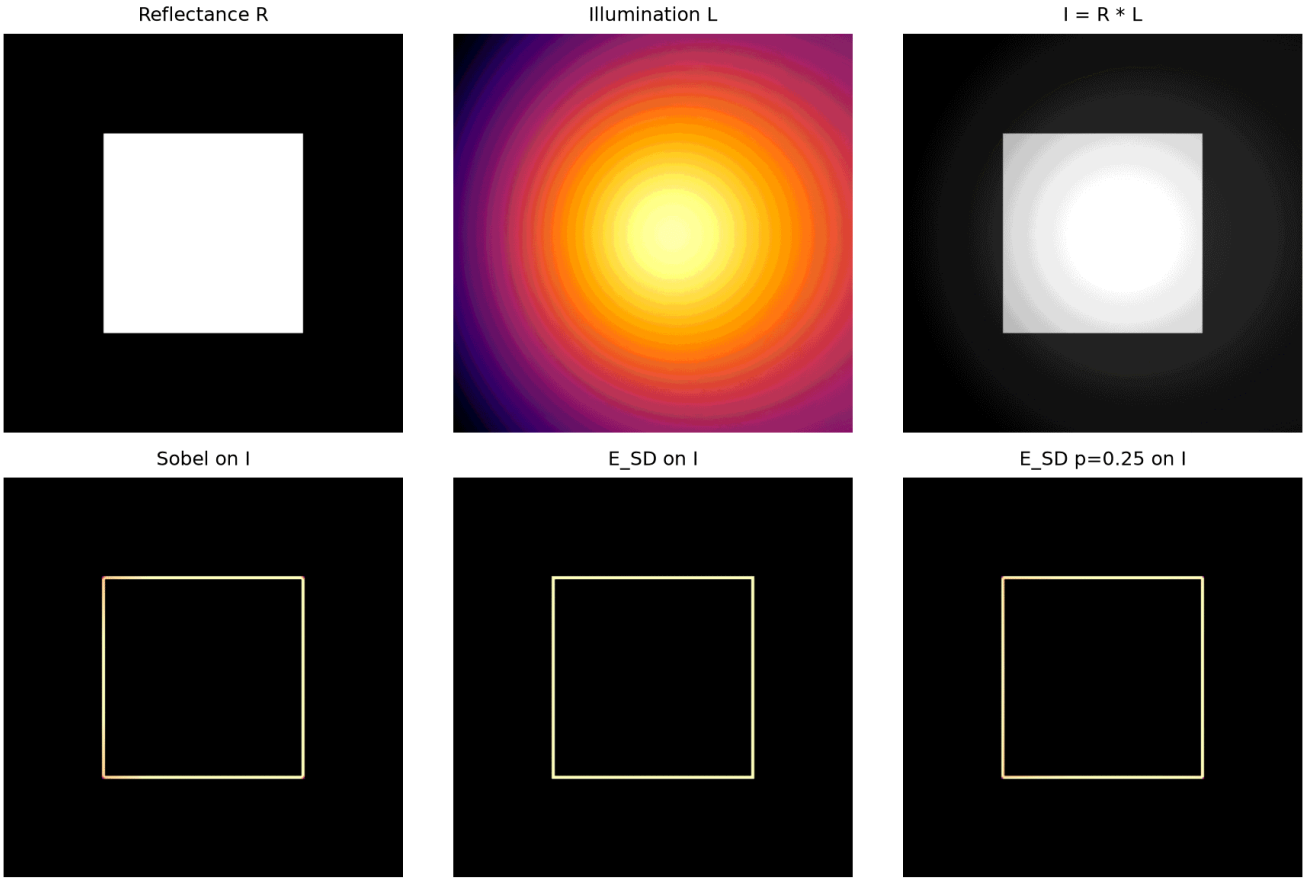


Fig. 11: Controlled checks of the relative change on analytic images. Top row: reflectance R, smooth illumination field L and the image $I = R \cdot L$; bottom row: Sobel, E_{SD} ($p = 1$) and $E_{0.25}$ on I, all scaled to their 99th percentile. Sobel also responds to the illumination gradient, whereas the relative operators are close to the reflectance edges.

$p \approx 0.25$ and a loss at $p = 1$ — not a new state of the art in boundary detection. The natural next step is to embed E_p in a full edge pipeline with smoothing, non-maximum suppression and hysteresis thresholding and to re-evaluate whether the same interior exponent is preferred.

Data and code availability

The BSDS500 dataset is publicly available from the Berkeley Computer Vision Group. The reference implementation, the synthetic evaluation, and the official BSDS500 boundary evaluation (`src/official_bsds/`), which links the unmodified Berkeley matching code, together with the frozen results reported in this paper, are available at `https://github.com/agarnung/computervisionmiscellaneous` (directory `papers/superderivative`). The frozen result files `src/synthetic_results.json`, `src/bsds_results_official_code.json` (validation split), `src/bsds_results_official_code_test.json` (held-out test split) and `src/bsds_results_official_code_percentile.json` (percentile normalisation), which the evaluation and the synthetic benchmark regenerate deterministically.

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